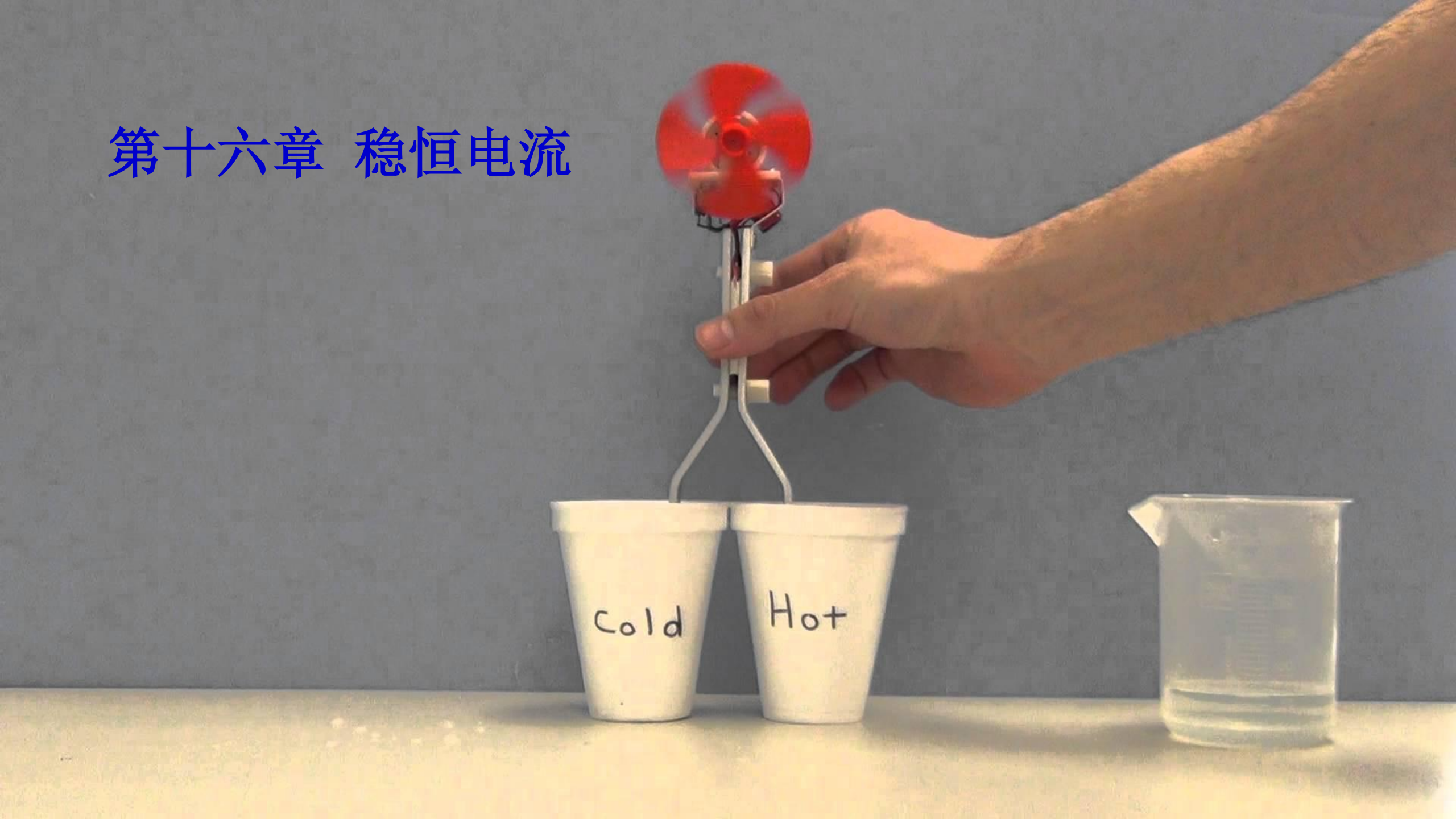
第十六章 稳恒电流



➢核心概念

- 电流密度
- 电动势

➢核心问题

- 什么是稳恒电流？
- 电流回路中满足什么定律？
- 什么驱动了电路回路中的电流？
- 电流的微观图像是什么？
- 电容器的充放电过程满足的方程是什么？

第十六章 稳恒电流 (steady current)

16.1 电流和电流密度

16.2 稳恒电流和稳恒电场

16.3 电动势

16.4 欧姆定律及其回路电压定律

16.5 电流的微观图象

16.6 电容器充放电

16.1 电流和电流密度

一. 电流强度

大小：单位时间内通过某一横截面的电量

$$\boxed{I = \frac{dq}{dt}}$$

方向：正电荷运动的方向

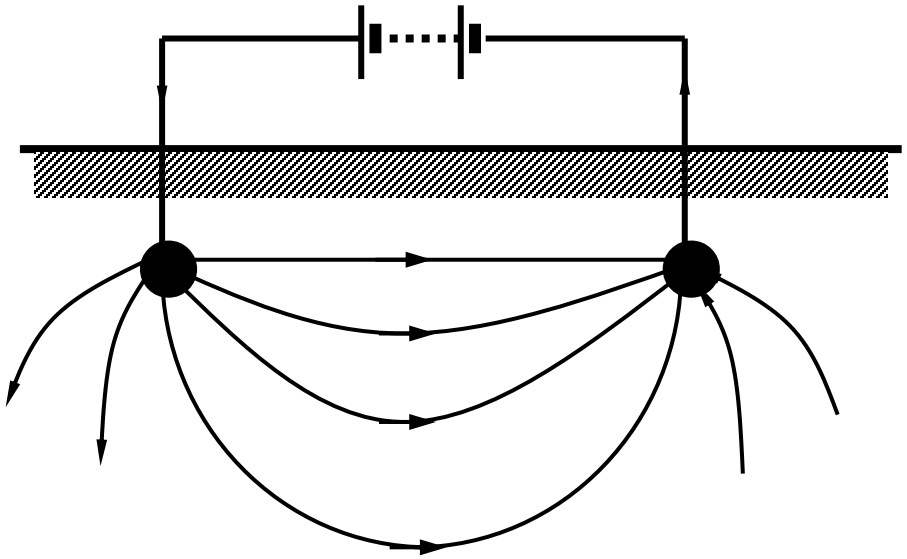
单位：安培

二. 电流密度 (current density)

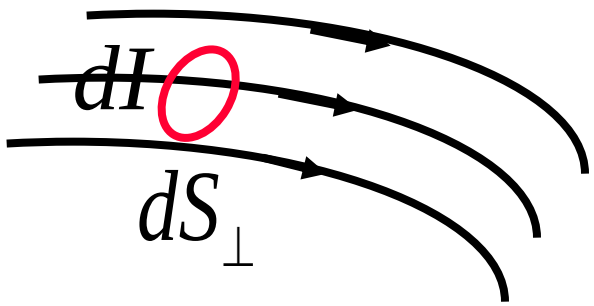
对大块导体，需电流密度的概念来进一步描写电流的分布。

例如：电阻法探矿

（图示）



$$\boxed{j = \frac{dI}{dS_{\perp}}}$$



电流密度:

正电荷定向移动方向垂直的单位面积上的电流强度

方向沿正电荷定向移动的方向

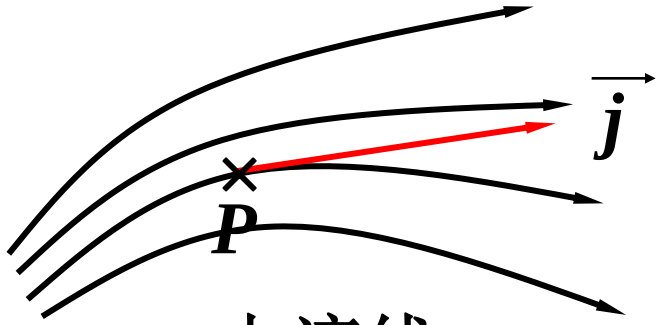
三. 电流线

“电流线”形象描写电流分布

电流线上某点的切向

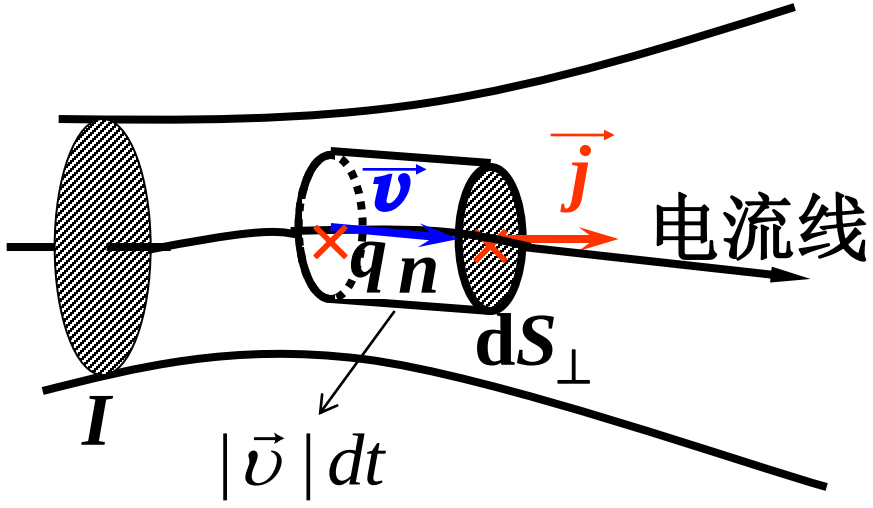
与该点 $\vec{j}$ 的方向一致;

电流线密的地方电流强



电流线

四. 电流密度和电流强度的关系



$$dq = nq|\vec{v}|dtdS_\perp$$

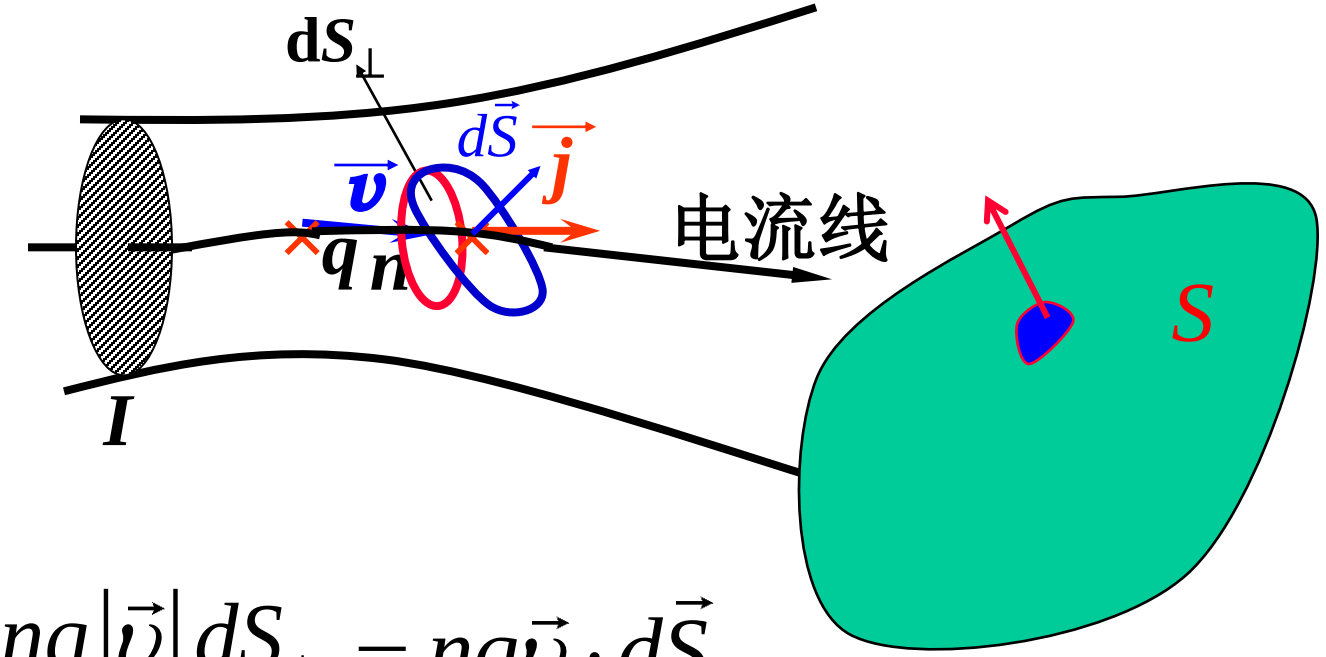
$$dI = nq|\vec{v}|dS_\perp$$

$$\boxed{\vec{j} = n\,q\,\vec{v}}$$

$\vec{v}$ —载流子定向移动速度

q — 载流子电荷

n— 载流子的浓度



$$dI = nq\left|\vec{\upsilon}\right| dS_{\perp} = nq\vec{\upsilon}\cdot d\vec{S}$$

$$\boxed{dI = \vec{j}\cdot d\vec{S}}$$

$$\boxed{I = \iint_{s} \vec{j}\cdot d\vec{S}}$$

对Cu： $j = 1\,\mathrm{A/mm^2}$ 时， $\boldsymbol{v} \approx 7.4\times10^{-2}\,\mathbf{mm/s}$

PRL **96,** 037601 (2006) PHYSICAL REVIEW LETTERS week ending 27 JANUARY 2006

漂移速度的测量

PRL 96, 037601(2006).

Conduction-Electron Drift Velocity Measurement via Electron Spin Resonance

M. Drescher,[1,*] N. Kaplan,[2] and E. Dormann[1]
[1]*Physikalisches Institut, Universität Karlsruhe(TH) D-76128 Karlsruhe, Germany*[†]
[2]*Racah Institute of Physics, Hebrew University of Jerusalem, 91904 Jerusalem, Israel*
(Received 8 September 2005; published 23 January 2006)

In analogy with NMR, motion induced phase shift of pulsed ESR signals enables in principle the direct detection of electron drift velocity or electronic current, respectively. Overcoming the difficulties with additional magnetic field gradients induced by the current itself, we succeeded in demonstrating the detection of electron flow via ESR. Measuring the electron drift velocity in the organic conductor (fluoranthene)$_2$PF$_6$ the microscopic Ohmic law could be observed in a current range of more than ± 0.25 A.

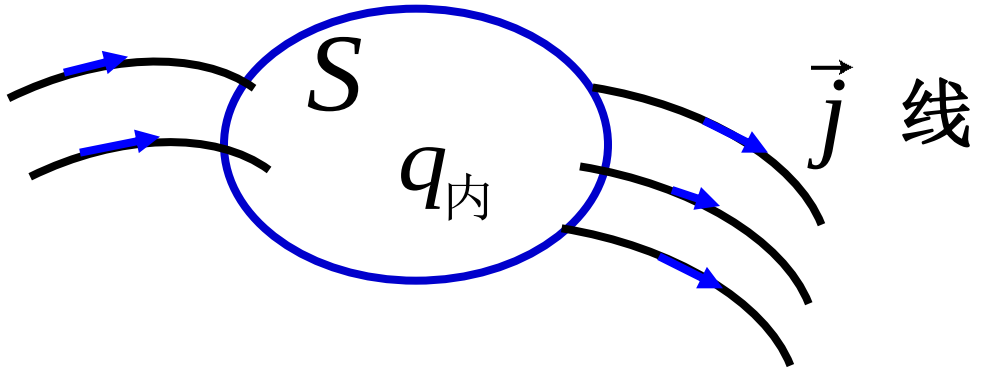
DOI: 10.1103/PhysRevLett.96.037601 PACS numbers: 76.30.Pk, 72.80.Le, 76.60.Pc

电流有热效应，故应限制 j 的大小

例如对**Cu**导线要求： $j \sim 10\,\mathrm{A/mm^2}$

对于超导导线， j 可达$\mathbf{10^4 A/mm^2}$

五. 电流连续性方程



$$\boxed{\oiint_S \vec{j}\cdot d\vec{S} = -\frac{dq_{内}}{dt}}$$

电流线发出于正电荷减少的地方

终止于正电荷增加的地方

16.2 稳恒电流和稳恒电场

一. 稳恒电流

电流场中每一点的电流密度的大小和方向均不随时间改变

二. 稳恒条件

1. 稳恒条件

$$\oiint_S \vec{j}\cdot d\vec{S} = -\frac{dq_{内}}{dt}$$

稳恒条件下闭合曲面积分为常数，而且一定为零。

$$\boxed{\oiint_S \vec{j}\cdot d\vec{S} = 0}$$

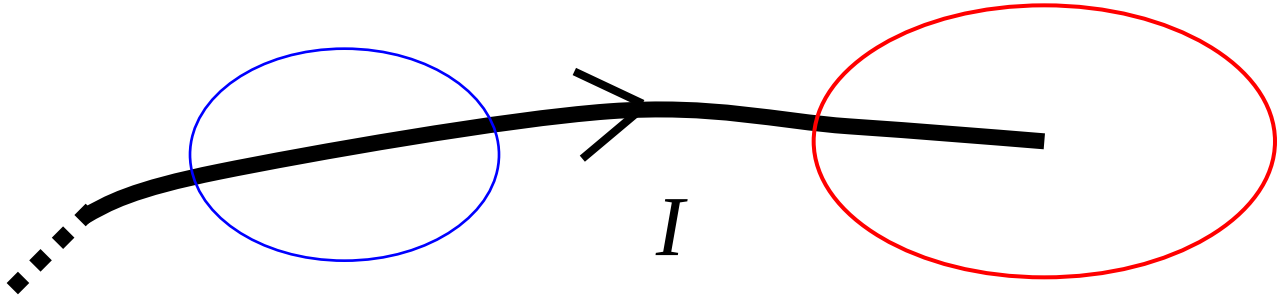
$$\frac{dq_{内}}{dt} = 0 \qquad \rho = \text{const.}$$

电荷分布不变

2.由稳恒条件可得出几个结论

- 稳恒电流的电路必须闭合

$$\oiint_S \vec{j}\cdot d\vec{S}=0$$

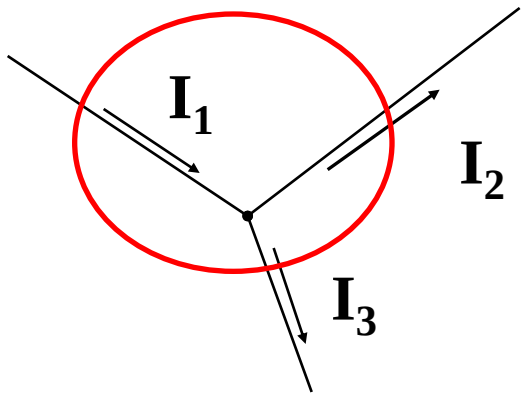


- 导体表面电流密度矢量无法向分量

- 对一段无分支的稳恒电路 其各横截面的电流强度相等

• 在电路的任一节点处 流入的电流强度之和等于流出节点的电流强度之和

--- 节点电流定律(基尔霍夫第一定律)



$$\oiint_S \vec{j} \cdot d\vec{S} = 0$$

$$I_1 = I_2 + I_3$$

三. 稳恒电场(本质是静电场)

•对于稳恒电路，导体内存在不随时间改变的电荷分布，由此产生稳恒电场

电场性质

- 电荷分布以及电场不随时间改变
- 满足高斯定理
- 满足环路定理, 是保守场, 可引入电势差

导体处于非静电平衡

- 稳恒电流的电场总是伴随电荷的定向移动，导体内恒定电场不为零
- 稳恒电场对定向移动电荷作功，需要提供能量维持稳恒电场

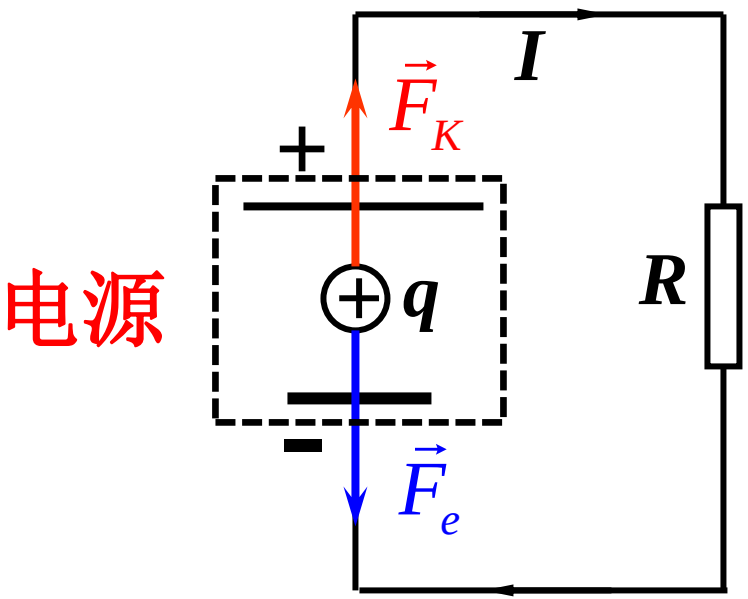
16.3 电动势 (electromotive force)

$$q \Rightarrow \vec{E} \Rightarrow I$$

稳恒电流，电路必须闭合

$$\oint_L \vec{E} \cdot d\vec{l} = 0 \Rightarrow \oint_L q\vec{E} \cdot d\vec{l} = 0$$

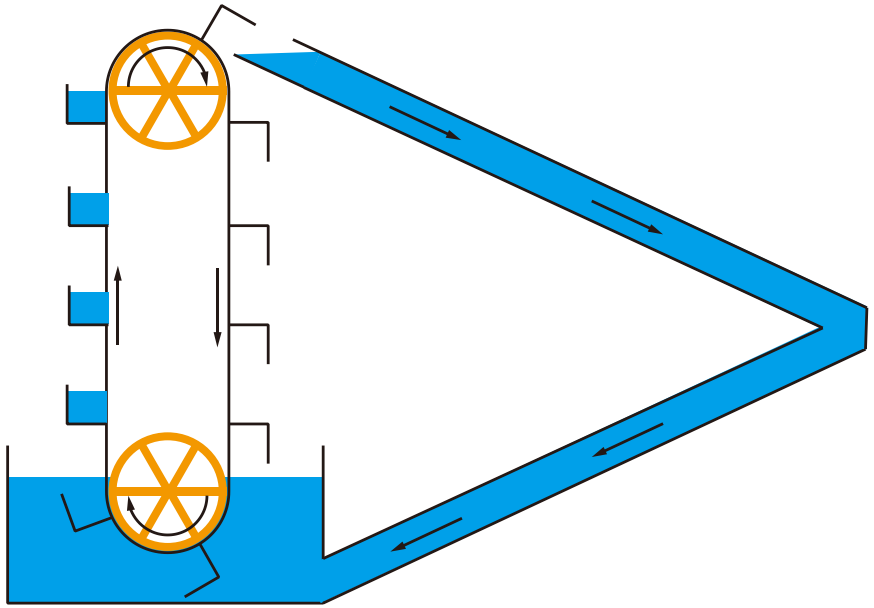
必须有非静电力$\vec{F}_K$存在才能在闭合电路中形成稳恒电流



非静电力场强 $\boxed{\vec{E}_K = \dfrac{\vec{F}_K}{q}}$

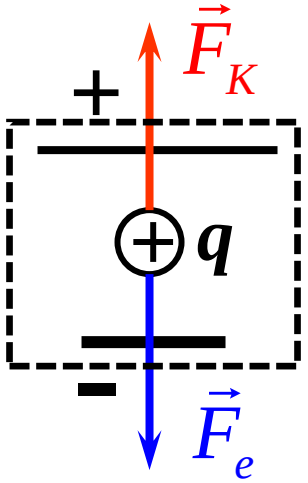
电动势：

把单位正电荷经电源内部由负极移向正极过程中非静电力所作的功



$$\boxed{\varepsilon = \int_{-}^{+} \vec{E}_K \cdot d\vec{l} = \oint_L \vec{E}_K \cdot d\vec{l}}$$

电源



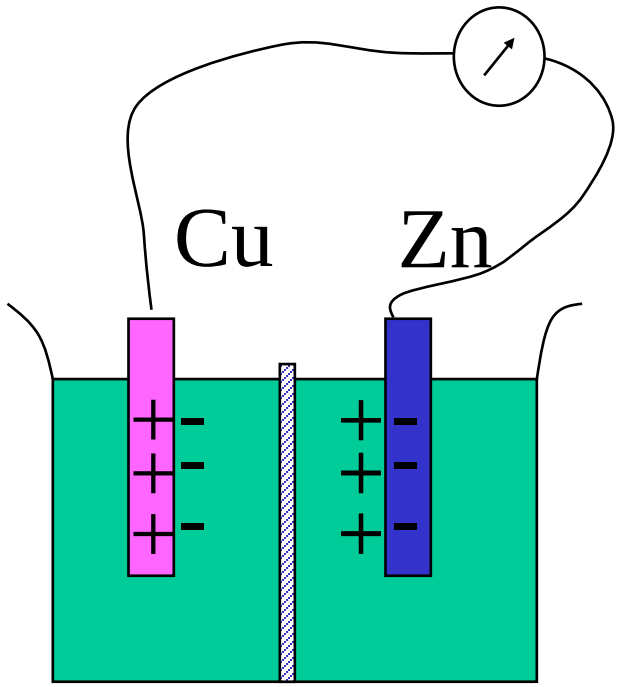
$$\vec{E}_K + \vec{E}_e = 0$$

$$\varepsilon = \int_-^+ \vec{E}_K \cdot d\vec{l} = -\int_-^+ \vec{E}_e \cdot d\vec{l} = U$$

有各种不同的电源：电磁、化学、热、光等。

化学电池原理

丹聂耳电池



$Cu^{++} \rightarrow Cu$

$Zn \rightarrow Zn^{++}$

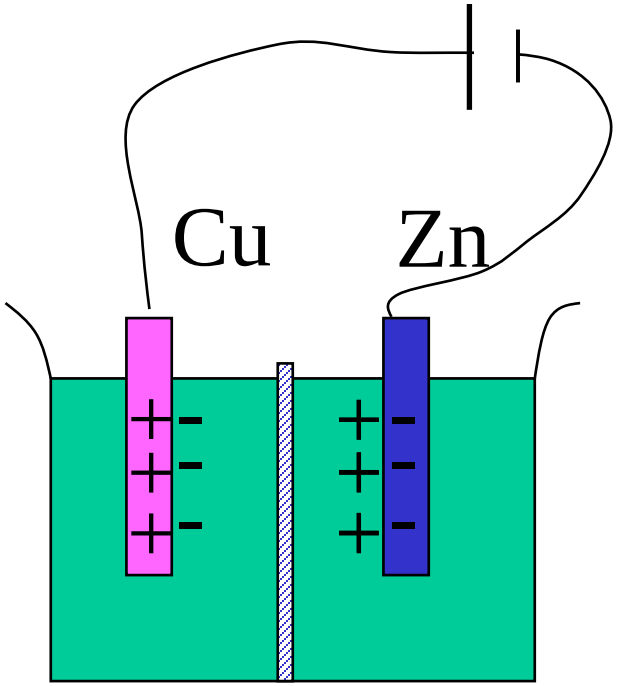
$CuSO_4$ $ZnSO_4$

U_{Cu}

$U_{溶液}$

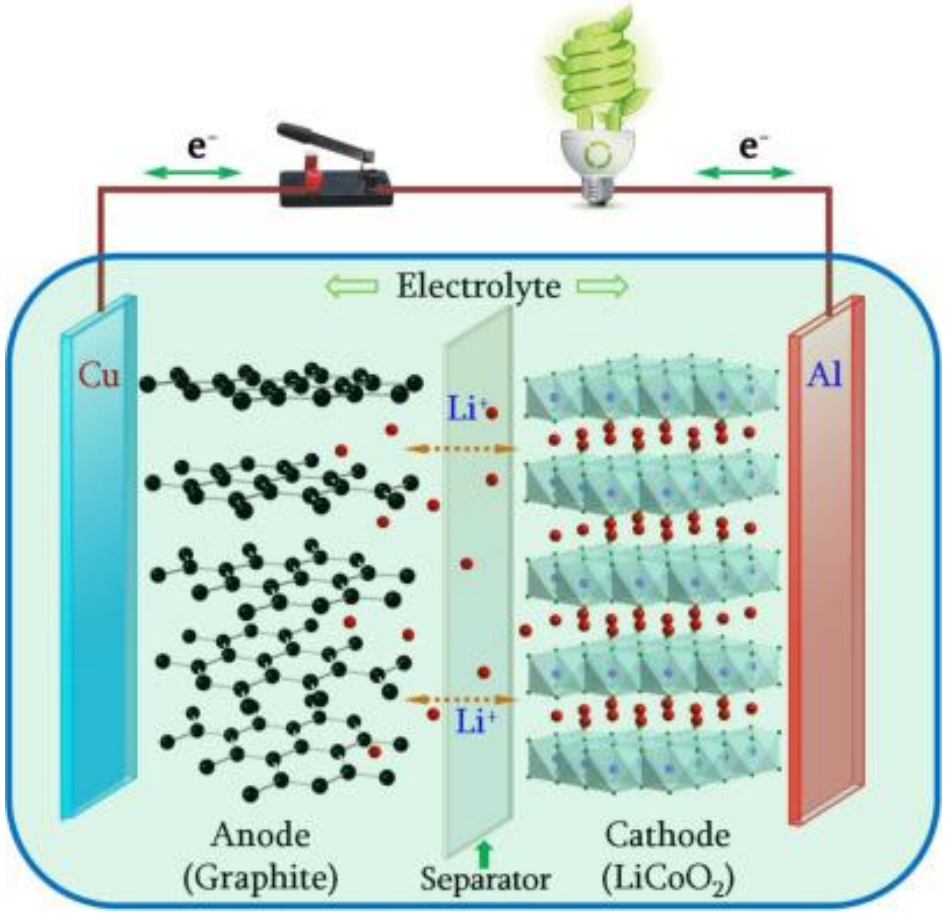
U_{Zn}

充电是逆过程



$CuSO_4$ $ZnSO_4$

锂离子电池工作原理*



正极反应

$$LiMO_2 \underset{Discharge}{\overset{Charge}{\rightleftarrows}} Li_{1-x}MO_2 + xLi^+ + xe$$

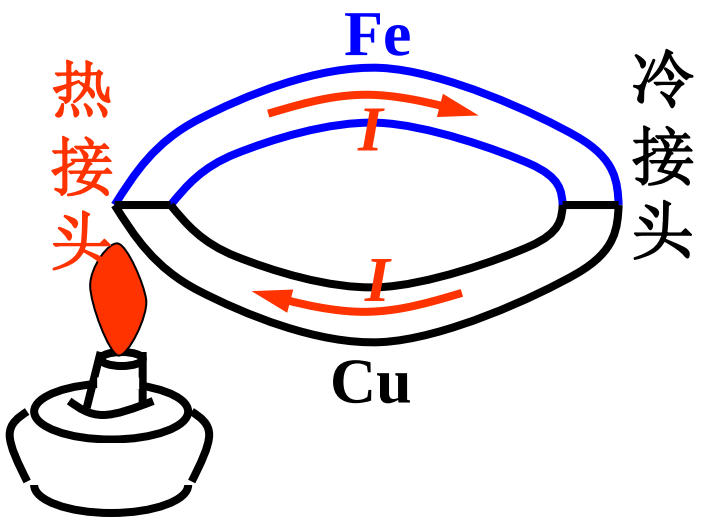
负极反应

$$nC + xLi^+ + xe \underset{Discharge}{\overset{Charge}{\rightleftarrows}} Li_xC_n$$

电池总反应

$$LiMO_2 + nC \underset{Discharge}{\overset{Charge}{\rightleftarrows}} Li_{1-x}MO_2 + Li_xC_n$$

* 温差电现象



回路中形成温差电动势

（席贝克[Seebeck]效应，1821）

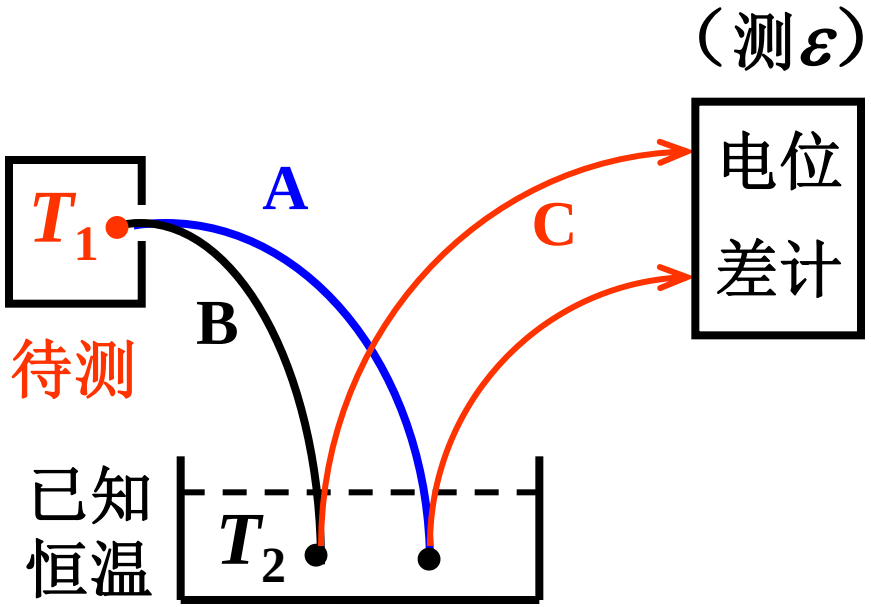
（演示实验）

温差电动势的成因：

▲在同种金属中，温差形成自由电子的热扩散（汤姆孙[Thomon]电动势）

▲在不同金属中，因自由电子浓度不同，在接头处产生与温度有关的扩散（珀耳帖[Peltier]电动势）

热电偶：



一般 ε~mV/100 ℃

Bi – Sb ε~10^{-2}V/100 ℃

（铋）（锑）

优点：

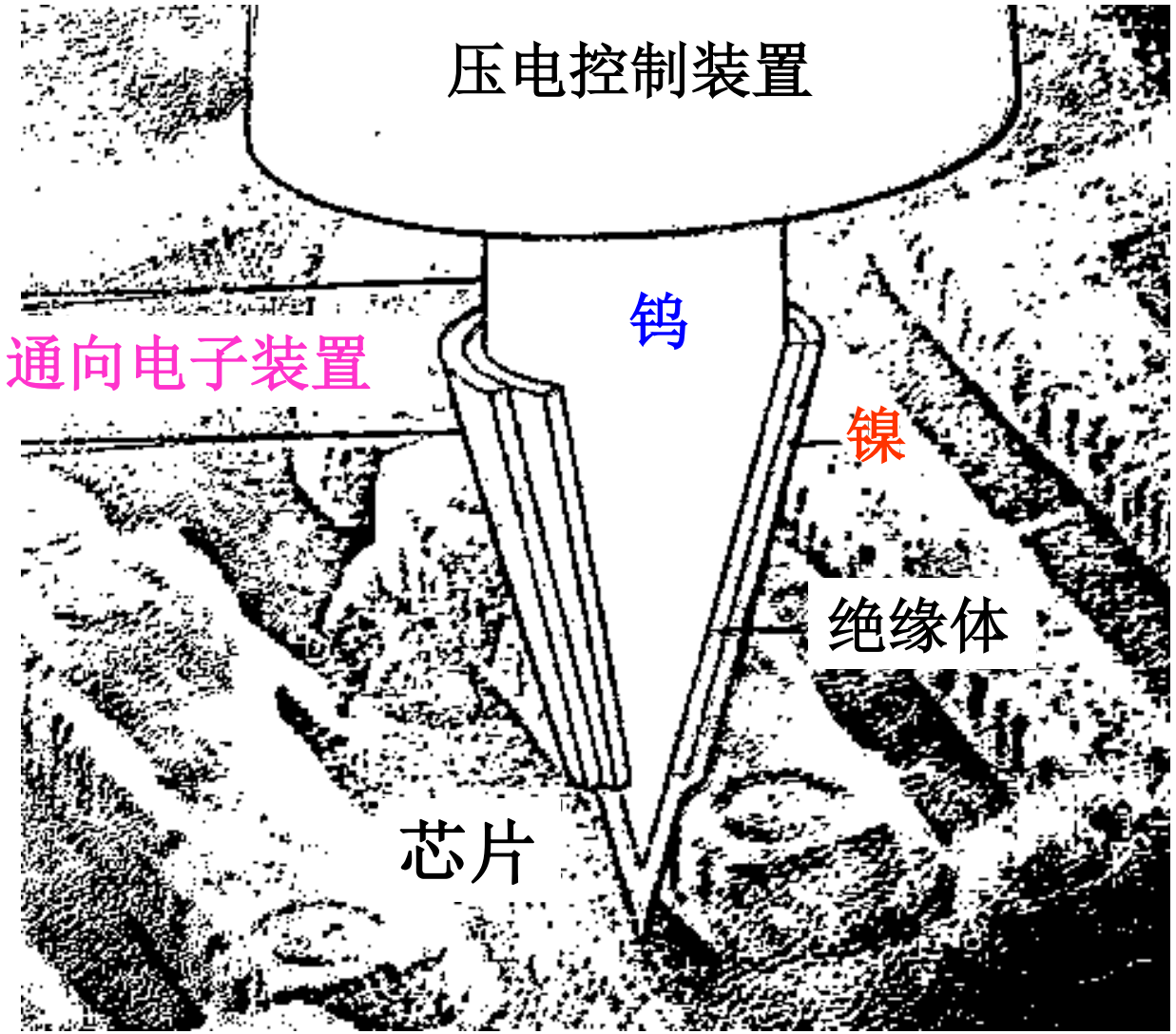
▲热容小，灵敏度高（10^{-3} ℃）

▲测温范围大（–200 ℃ — 2000 ℃）

▲可逐点测量，测小范围内温度变化

▲便于自动控制

温差电现象的现代应用实例：



扫描热显微镜

钨和镍在探针尖处相接，形成热电偶测温端

扫描热显微镜

▲性能：热探针针尖直径只有约30nm，可在数十纳米的尺度上，测出万分之一度的温度变化。

▲工作原理：通电流使探针加热并接近试样表面。

针尖和被测表面距离↓→针尖散热↑→温度↓；

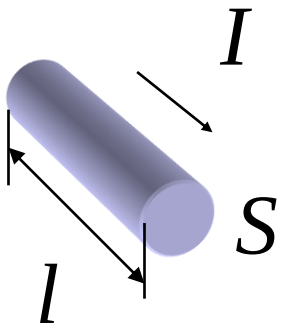
针尖和被测表面距离↑ →针尖散热↓→温度↑。

当探针在试样表面上扫描时，其温度变化代表试样表面的起伏状况。

16.4 欧姆定律及其回路电压定律

一. 欧姆定律微分形式

欧姆定律



$$\boxed{R = \frac{U}{I}}$$

$$\boxed{R = \rho \frac{l}{S}}$$

电阻率

$$I = jS \qquad U = El$$

$$E = \rho j = \frac{j}{\sigma}$$

电导率

考虑方向 $\boxed{\vec{E} = \frac{\vec{j}}{\sigma}}$

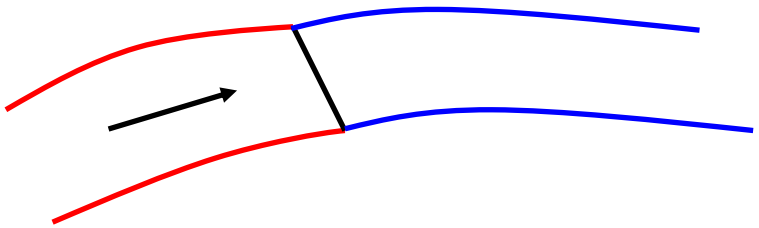
$$\boxed{\vec{j} = \sigma \vec{E}}$$

电流是电场驱动下的电荷定向移动

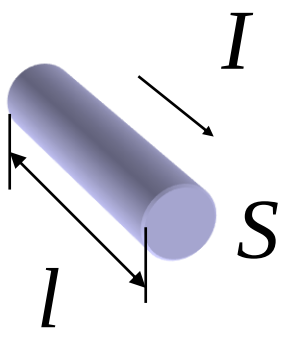
由于稳恒电流 $$\boxed{\oiint_S \vec{j} \cdot d\vec{S} = 0}$$

均匀导体内： $$\boxed{\oiint_S \vec{E} \cdot d\vec{S} = 0}$$

稳恒电流导线上电荷分布在表面或界面



- 焦耳定律
(单位体积电功率)



$$p = IU$$

$$= jSEl$$

$$= j\frac{j}{\sigma}V$$

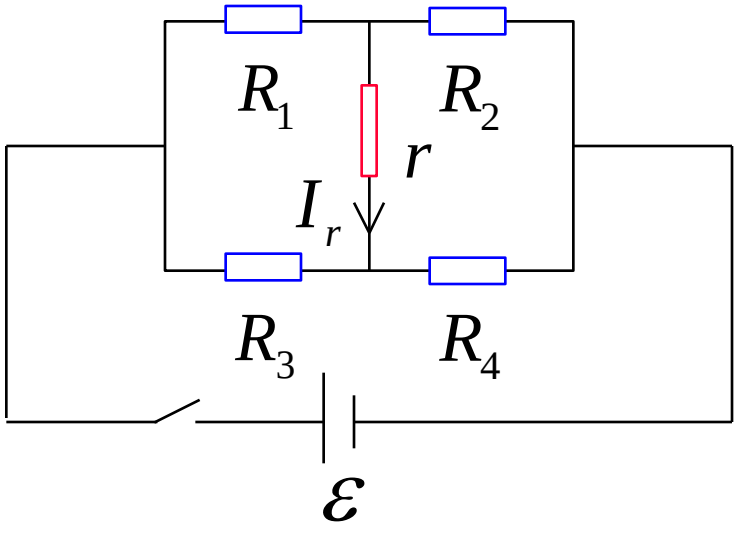
$$\boxed{p = j^2/\sigma}$$

二. 有电动势时欧姆定律

非静电场与静电场对电荷的作用是等效的

$$\boxed{\vec{j} = \sigma(\vec{E} + \vec{E}_K)}$$

三. 基尔霍夫第二定律



串并联公式不能解决

基尔霍夫第一定律（稳恒电流条件）

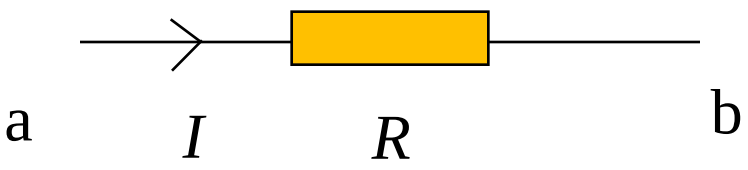
稳恒电场的一个基本性质满足环路定理，是保守场

$$\boxed{\oint_L \vec{E}\cdot d\vec{l} = 0}$$

在稳恒电路中沿任何闭合回路一周的电势降落的代数和等于零。

回路电压定律(基尔霍夫第二定律)

电路中如何表示?

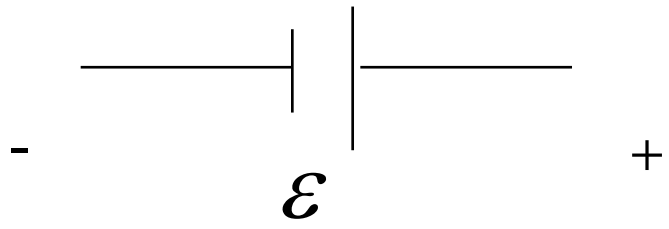


$$\int_a^b \vec{E}\cdot d\vec{l} = \int_a^b \frac{\vec{j}}{\sigma}\cdot d\vec{l}$$

$$= \int_a^b Sj\rho\frac{dl}{S} = I\int_a^b \rho\frac{dl}{S} = IR$$

积分沿电流的方向，电阻上电压降IR

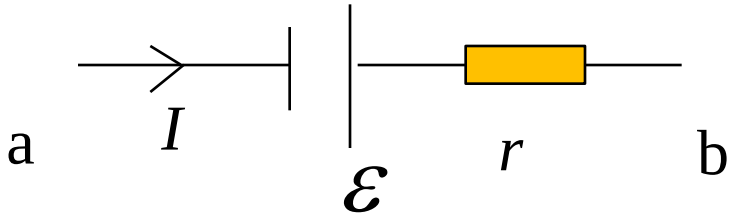
否则$-IR$



$$\int_{-}^{+} \vec{E}_K \cdot d\vec{l} = \varepsilon$$

非静电场积分从负极到正极 ε

否则 $-\varepsilon$



$$\vec{j} = \sigma(\vec{E} + \vec{E}_K)$$

$$V_{ab} = \int_a^b \vec{E} \cdot d\vec{l} = \int_a^b \left(\frac{\vec{j}}{\sigma} - \vec{E}_K\right) \cdot d\vec{l} = Ir - \varepsilon$$

基尔霍夫第二定律

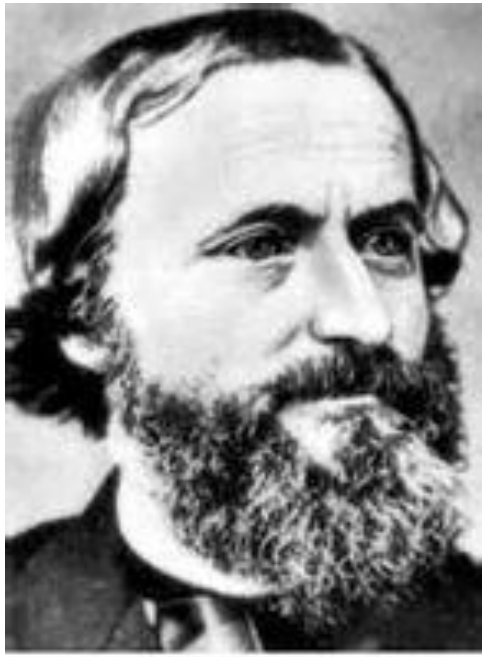
$$\oint_L \vec{E}\cdot d\vec{l} = 0$$

$$\vec{j} = \sigma(\vec{E} + \vec{E}_K)$$

$$\oint_L \left(\frac{\vec{j}}{\sigma} - \vec{E}_K\right)\cdot d\vec{l} = 0$$

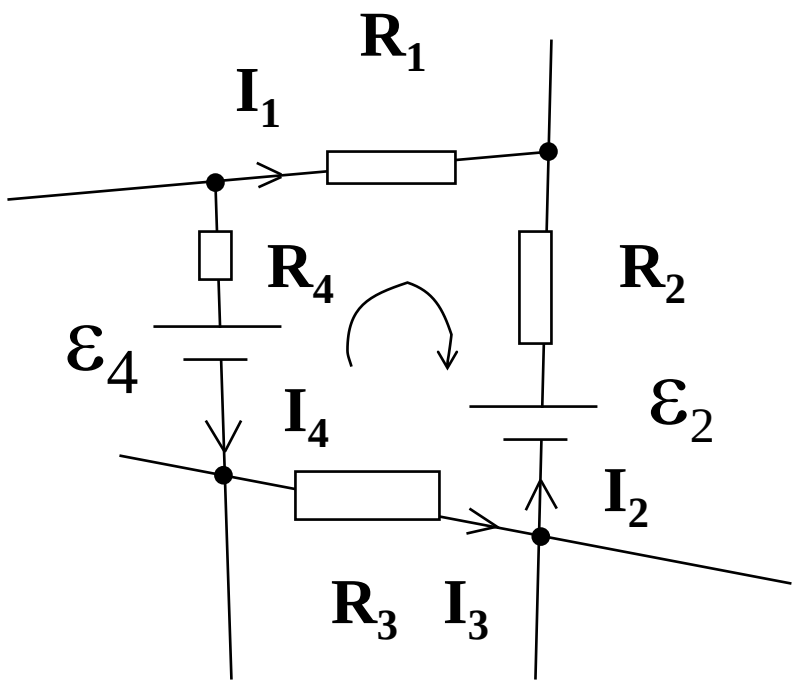
$$\boxed{\sum(\pm)IR + \sum(\pm)\varepsilon = 0}$$

I_i与L绕向一致为正，
ε_i正极到负极为正



基尔霍夫，G.R.

例1 求下图回路电压表示

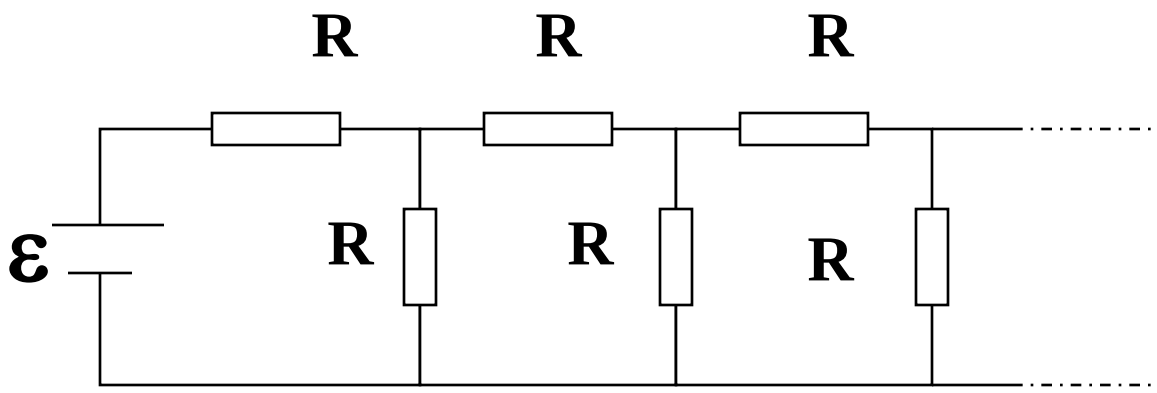


解:

$$I_1R_1 - I_2R_2 - I_3R_3 - I_4R_4 + \varepsilon_2 - \varepsilon_4 = 0$$

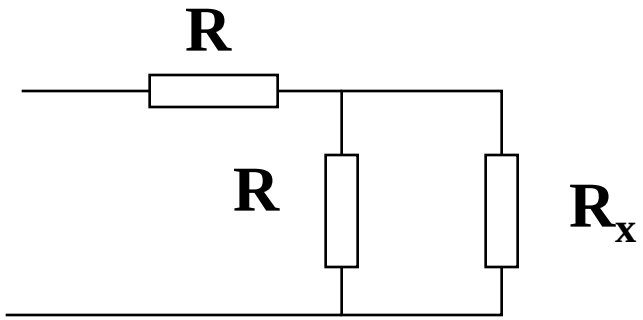
I_i 与 L 绕向一致为正，ε_i 正极到负极为正

例2



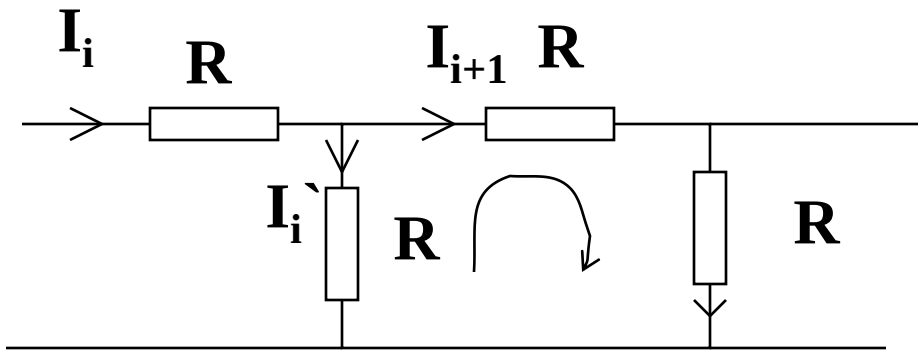
求：1）等效电阻；2）第n个电阻上电流

解：1)



$$R_x = R + \frac{R_x R}{R + R_x}$$

$$R_x = \frac{1+\sqrt{5}}{2} R$$



2)

$$I_i = I_{i+1} + I'_i$$

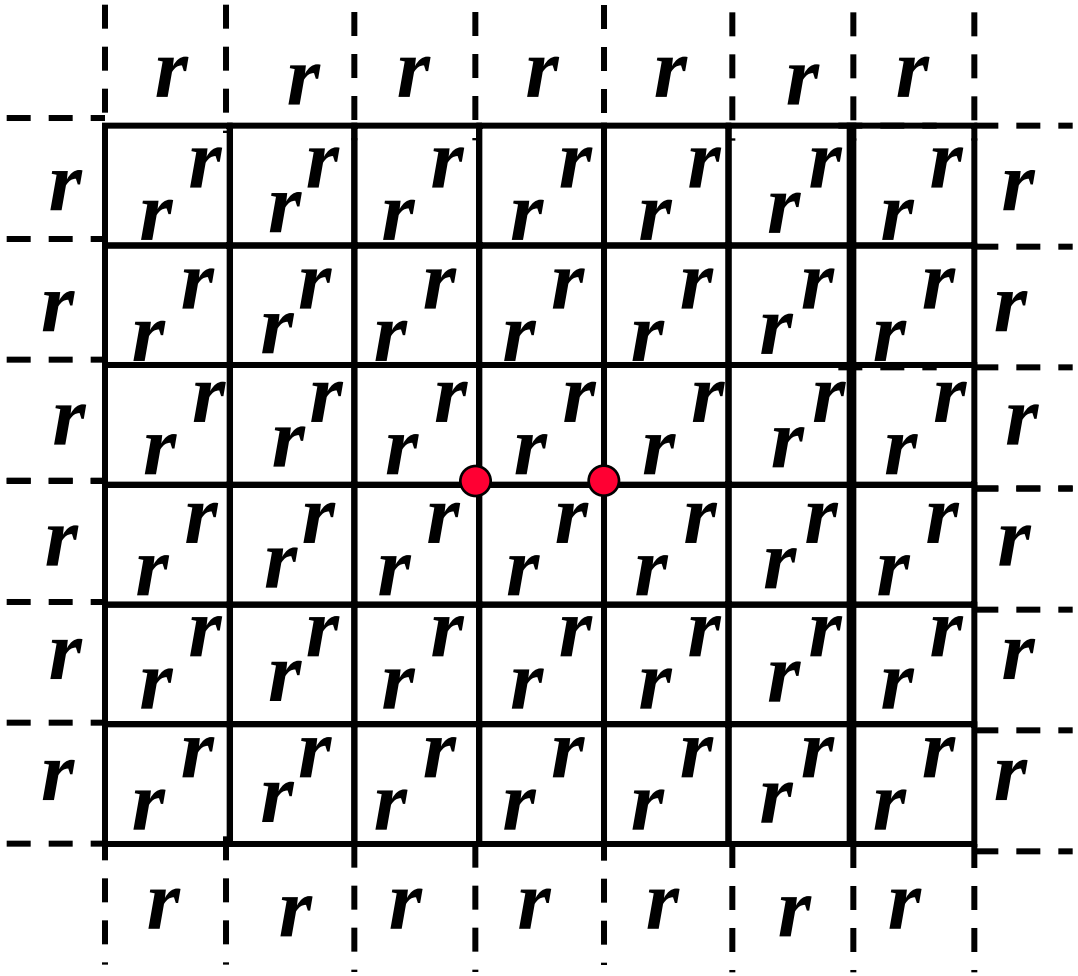
$$I_{i+1} R_x - I'_i R = 0$$

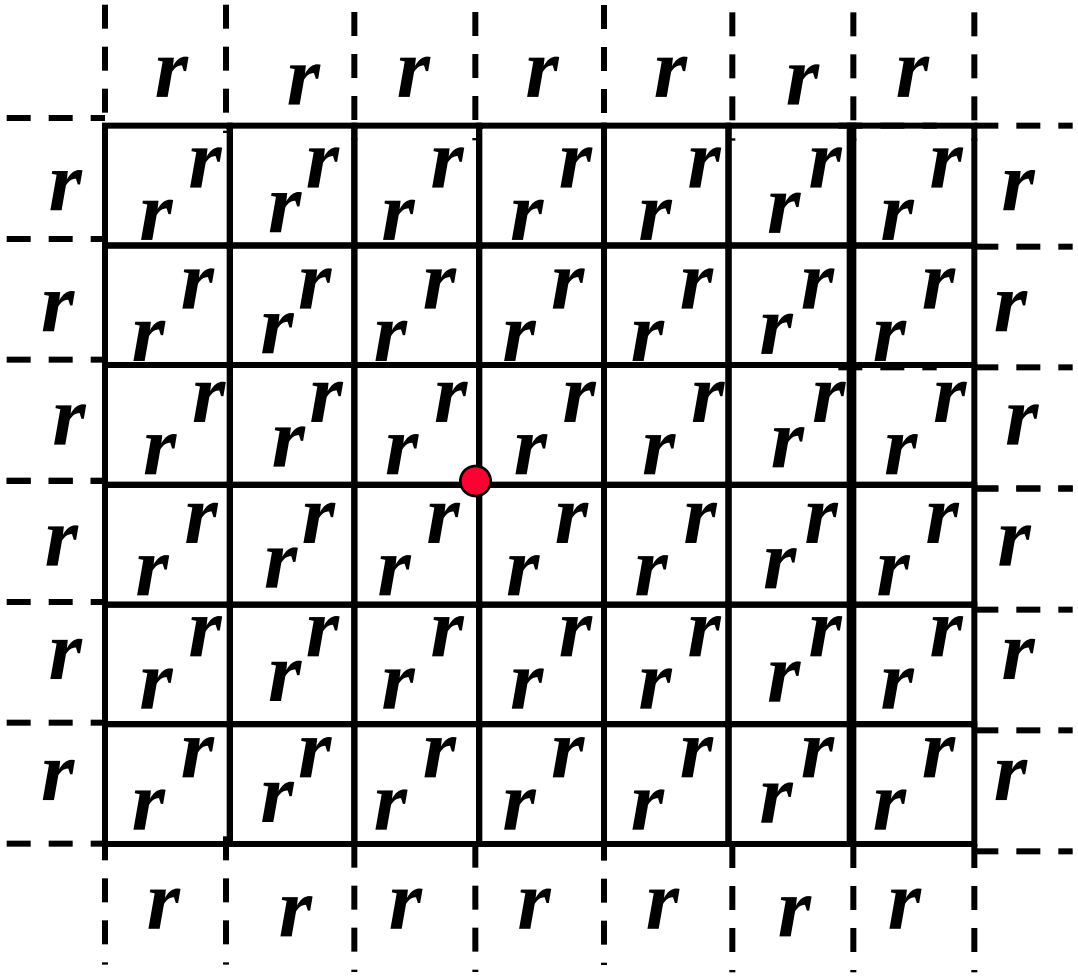
$$I_{i+1} = \frac{R}{R + R_x} I_i \qquad I'_i = \frac{R_x}{R + R_x} I_i \qquad I_1 = \frac{\varepsilon}{R_x}$$

$$I_n = \left(\frac{R}{R + R_x}\right)^{n-1} I_1 = \left(\frac{2}{1+\sqrt{5}}\right)^{2n-1} \frac{\varepsilon}{R}$$

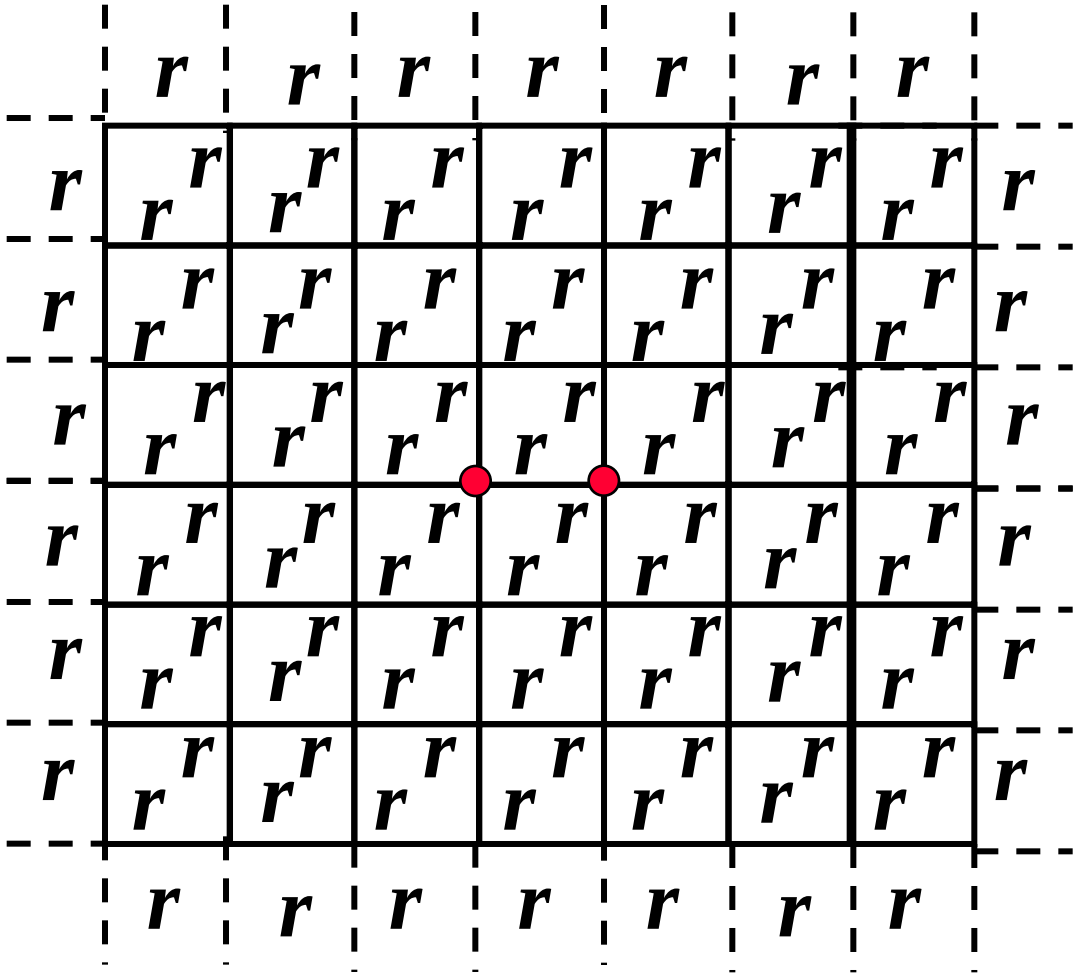
$$I'_n = \left(\frac{2}{1+\sqrt{5}}\right)^{2n} \frac{\varepsilon}{R}$$

例3 求等效电阻



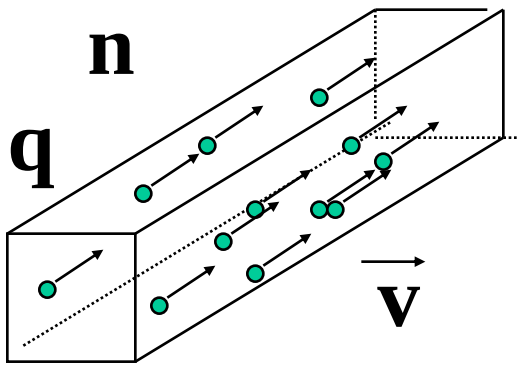


$$I = \frac{1}{4} I_0$$



解: $$\frac{I}{4}\times 2\cdot r = IR_x \qquad R_x = \frac{r}{2}$$

16.5 电流的微观图象



$$\vec{j} = nq < \vec{v} >$$

$$\vec{v} = \vec{v}_0 + \frac{q\vec{E}}{m}t$$

取平均 $\langle \vec{v}_0 \rangle = 0$ $\quad \langle t \rangle = \tau$ $\quad \langle \vec{v} \rangle = \frac{q\vec{E}}{m}\tau$

$$\vec{j} = \frac{nq^2\tau\vec{E}}{m} = \sigma\vec{E}$$

$$\boxed{\sigma = \frac{nq^2\tau}{m}}$$

$$\vec{j} = \sigma \vec{E}$$

$$\boxed{\sigma = \frac{nq^2\tau}{m}}$$

失效？

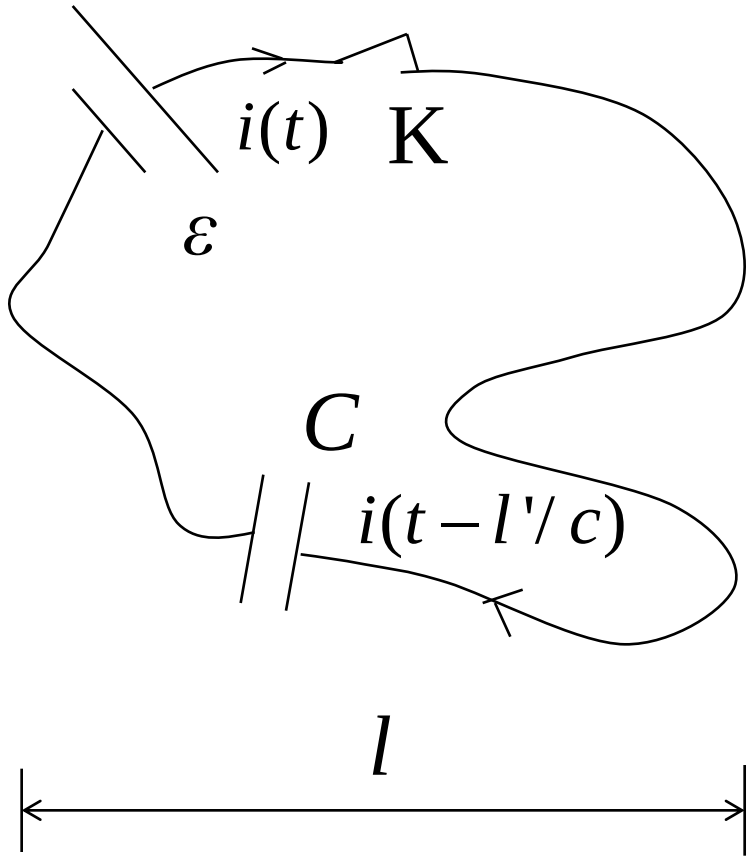
1 平均电流密度 有涨落 热噪音；
2 电场太强 平均碰撞时间受影响 失效；
3 电场变化时间与平均碰撞时间可比拟失效；
4 低温时量子效应明显 失效。

电阻率 $$\boxed{\rho = \frac{m}{nq^2}\frac{1}{\tau}}$$

$$\boxed{\frac{1}{\tau} = \frac{1}{\tau_{e-imp}} + \frac{1}{\tau_{e-e}} + \frac{1}{\tau_{e-ph}} + \ldots\ldots}$$

电阻随温度增长, ……

16.6 电容器充放电(暂态过程)*



似稳电路近似

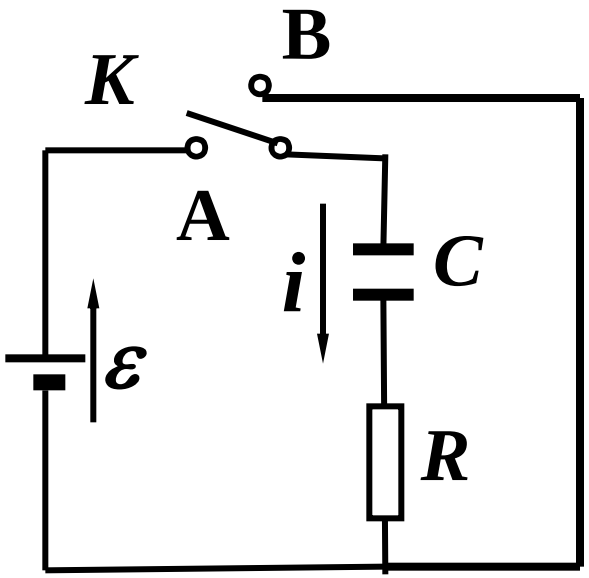
电路特征时间 $\tau >> l/c$

$$i(t-l'/c)\approx i(t) \qquad l'/c << \tau$$

交流电似稳近似:

$c/f >> l$

注意：电容处基尔霍夫第一定律不能用



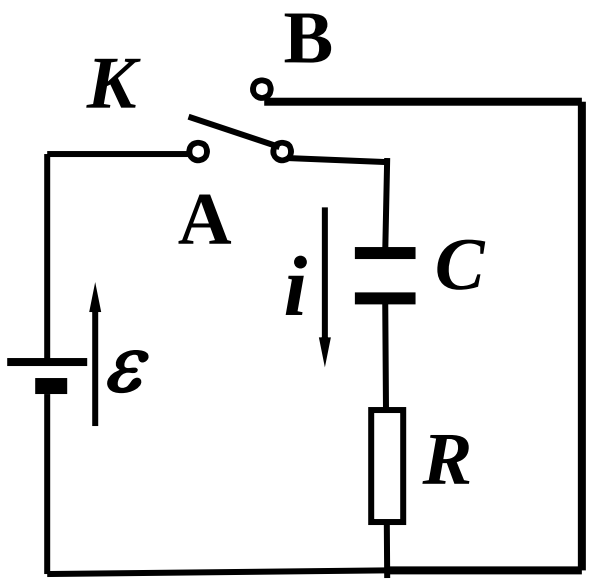
(K-A) $$-\varepsilon + iR + \frac{q}{C} = 0$$

(K-B) $$iR + \frac{q}{C} = 0$$

$$i = \frac{dq}{dt}$$

$$\frac{dq}{dt} + \frac{q}{CR} = \frac{\varepsilon}{R} \xrightarrow{q' = q - C\varepsilon} \frac{dq'}{dt} + \frac{q'}{CR} = 0$$

$$\frac{dq}{dt} + \frac{q}{CR} = 0 \qquad \tau = RC$$ ——特征时间



$$\frac{dq'}{dt} + \frac{q'}{\tau} = 0$$

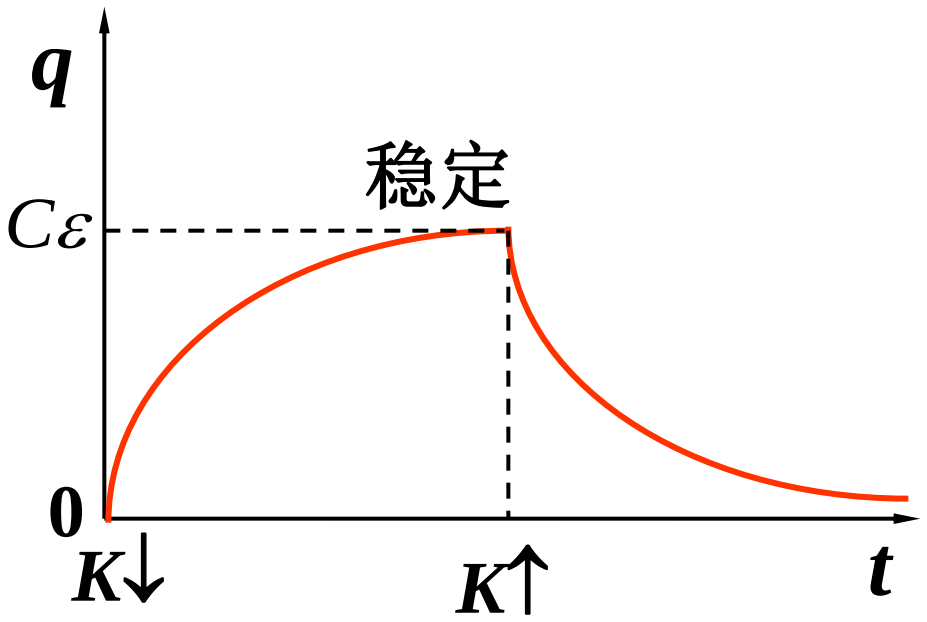
$$q' = Ae^{-\frac{t}{\tau}}$$

(K-A) $q|_{t=0} = 0$ $\quad q'|_{t=0} = -C\varepsilon$

$$q' = q - C\varepsilon \qquad q = C\varepsilon(1 - e^{-\frac{t}{\tau}})$$

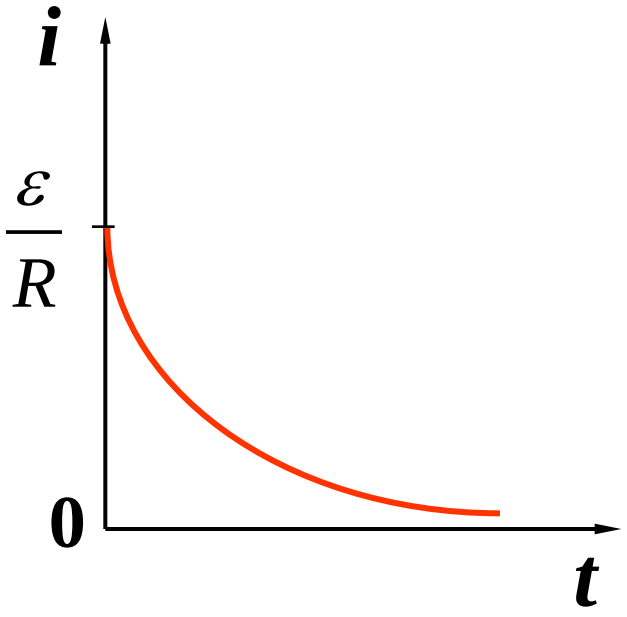
$$q' = -C\varepsilon e^{-\frac{t}{\tau}}$$

(K-B) $q|_{t=0} = C\varepsilon$ $\quad q = C\varepsilon e^{-\frac{t}{\tau}}$



$$q = C\varepsilon(1 - e^{-\frac{t}{\tau}})$$

$$q = C\varepsilon e^{-\frac{t}{\tau}}$$



$$i = \dot{q} = \frac{\varepsilon}{R} e^{-t/\tau}$$